

SHRI RAJ BISARIA

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SUMMARY

AI/ML engineer and researcher specializing in security systems, DL infrastructure, and optimizer theory. Solo author of an ICLR 2027 paper on spectral optimizer geometry (OpenReview #8176). Founder of EdgeCI (YC W27 applicant), an Apple Silicon inference regression gate adopted upstream by RapidMLX. Solo winner of the Trae AI Hackathon, global finalist in the Meta PyTorch OpenEnv Hackathon (solo, 70k+ teams), QuantCraft finalist, and Smart India Hackathon semi-finalist. Builds in C++, Go, Python, Rust, Swift, and TypeScript.

RESEARCH & PUBLICATIONS

- When Does Adaptive Spectral Normalization Define a Fixed Convex Geometry? / Solo Author** ICLR 2027 — Under Review
- Determined when spectral optimizers (**Muon**, **Newton-Schulz**, **Jordan polynomials**) admit a fixed convex potential. Proved one cubic Newton-Schulz step is norm-derived but repeated steps and all positive-depth Taylor-quintic compositions are not, despite preserving order.
 - Derived a sharp filter-preserving condition-number frontier. Verified all critical points via exact **FLINT/Arb** rational certificates over $\mathbb{Q}$ and dependent replay of 4,728 stored spectra. Registered abstract **#8176** on OpenReview.

FOUNDER EXPERIENCE

- EdgeCI — Founder & Core Engineer (YC W27 Applicant)** 2026 – Present
- Built a statistical performance-release gate for **llama.cpp** (121k+ ★) and MLX on Apple Silicon. Paired ABBA/BAAB scheduling with thermal preflight settling and log-ratio confidence intervals produce PASS/FAIL/INCONCLUSIVE verdicts on sub-5% regressions.
 - Established first M5 MacBook Air community benchmarks, cited and adopted upstream by **RapidMLX** as official hardware reference data. Collaborated directly with upstream open-source maintainers.

SELECTED PROJECTS

- Thinking Budget — Adaptive Reasoning for LLMs** Python, PyTorch, GRPO, TRL, Qwen3 1.7B, OpenEnv
- Spun out of DeathClock's CVE dataset: designed an RL environment teaching a 1.7B model dynamic reasoning effort allocation across 150 real CVEs. Multi-objective reward for calibration, difficulty awareness, and adversarial robustness (defeated 5/5 attack strategies).
 - Full **SFT → GRPO** pipeline pushed calibration accuracy 33% → 88% and triage F1 0.14 → 1.00. **Global Finalist (Solo), Meta PyTorch OpenEnv Hackathon 2026 (70,000+ teams)**. Live on HuggingFace Spaces.

- DeathClock — Predictive CVE Risk Platform** Python, XGBoost, LightGBM, React, D3.js, VS Code Extension, Gemini
- Solo-built platform scoring codebases for undisclosed CVE likelihood ahead of public disclosure: gradient-boosted models trained on NVD CVE data, interactive React/D3.js risk dashboard, VS Code extension, and Gemini plain-English explanations. Drew external documentation request from an **Apple PM**.

- godmode — Zero-Config Agent Prompt Compiler & CLI** TypeScript, Node.js, sqlite-vec, BM25, Ink/React TUI, NPM
- Published CLI (@godmode/cli) auto-generating configs for 11 coding agents (Claude Code, Cursor, Copilot). Built local hybrid BM25 + sqlite-vec vector search compressing context windows from 1.2M to 87K tokens (**93% reduction**). Includes prompt compiler and Claude pre-hook interceptors.

- drift — Zero-Entitlement Apple Neural Engine Monitor** Python, XNU Kernel (ctypes), IOKit, Textual TUI, SQLite
- htop-style ANE monitor for Apple Silicon: reads per-PID `ri_neural_footprint` and `ri_energy_nj` via `proc_pid_rusage` without root. Detects active neural models via FD scanning. 140-test suite (~91% coverage), multi-threaded TOPS benchmark, AppleSMC thermals. Distributed on **Homebrew**.

- PARASITE EVOLVED — Codebase Security Organism** Tree-sitter AST, LangGraph, Genetic Algorithms, Three.js
- Multi-agent security tool combining Tree-sitter AST parsing across 5 languages, genetic attack mutation search, and automated patch synthesis, visualized in an interactive 3D graph. Scans 2,000+ Java files in ~4s. **Finalist (Solo), QuantCraft Hackathon**.

- ZenTorrent — Open-Source CLI Streaming Engine** Go, BitTorrent Wire Protocol, Homebrew, Winget, Chrome Extension
- Streams torrents directly into VLC by prioritizing initial 5% piece chunks via an embedded HTTP server. Multi-threaded peer-wire parsing and piece verification. **90+ GitHub stars, 50K+ Reddit views, #1 Trending on r/golang & r/commandline**. Distributed via Homebrew, Winget, and Chrome extension.

- VOID Zero — Local Agent Control Plane** TypeScript, Node.js, MLX, Apple Silicon
- \$0-runtime local control plane translating Anthropic Messages API to local MLX/OpenAI endpoints on Apple Silicon with asynchronous SSE streaming, memory-pressure gating, and proposal → critique → action deliberation loops.

ACHIEVEMENTS & HONORS

- Hackathon Honors (Solo): Winner**, Trae AI Hackathon; **Global Finalist**, Meta PyTorch OpenEnv Hackathon 2026 (Thinking Budget, **70,000+ teams**); **Finalist**, QuantCraft Hackathon (PARASITE EVOLVED).
- Quantitative & Competitive Honors: IMC Prosperity Round 4** (delta-neutral options market maker, Black-Scholes, IV solver) & **Arbitrage Arena** (Pravega XII, IISc Bangalore).
- National Contests & Fellowships: Smart India Hackathon (SIH) Semi-Finalist** (ProjectPulse, MoSPI), **Adobe University Hackathon 2026** (R1 Coding/OA), **HackNova 2026 Rank 10 Finalist**, **IIT Madras AI/ML Excellence Program** (Microsoft Noida Campus cohort), **Google Solution Challenge 2026** (FairLens AI).
- Competitive Programming: LeetCode Rating 1808** (Top 8% globally), **Codeforces Pupil, AIR 18** DU Coding Contest.

TECHNICAL SKILLS

- Languages:** C/C++, Go, Python, Rust, Swift, TypeScript, JavaScript, Java, SQL
AI & Systems: PyTorch, GRPO, RLHF, Hugging Face (TRL), LangGraph, Metal, MLX, llama.cpp, XNU Kernel, IOKit, Lock-Free Atomics
Web & Tools: React, D3.js, Three.js, Git, Docker, Linux/POSIX, Homebrew (tap author), Winget, FLINT/Arb, SymPy

EDUCATION & CERTIFICATIONS

- Galgotias University — B.Tech in Computer Science and Engineering (Core)** 2024 – Present
Expected Graduation: 2028 | Greater Noida, India · **Certification:** DeepLearning.AI Specialization (Coursera / Andrew Ng)